ELECTRICITY BILL MANAGEMENT SYSTEM

1. Introduction

The Electricity Bill Management System is a Java-based console application developed using JDBC and MySQL. The purpose of this project is to automate the calculation of electricity bills and manage consumer billing records efficiently. This system reduces manual work, minimizes errors, and provides quick access to billing information.

The application follows a layered architecture using DTO, DAO, and Service layers, which makes the system modular, maintainable, and easy to understand. This project is developed as a college mini project.

2. Objectives of the Project

- To calculate electricity bills based on units consumed
- To store and manage electricity bill records in a database
- To perform CRUD operations (Create, Read, Update, Delete)
- To understand JDBC connectivity with MySQL
- To implement layered architecture in Java

3. Software Requirements

- Operating System: Windows 11
- Programming Language: Java (JDK 8 or above)
- Database: MySQL
- IDE: Eclipse
- JDBC Driver: MySQL Connector/J

4. Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Hard Disk: 20 GB free space

5. System Architecture

The project follows a layered architecture:

- Presentation Layer: MainApp.java (Console Interface)
- Service Layer: ElectricityBillService.java (Business Logic)
- DAO Layer: ElectricityBillDAO & ElectricityBillDAOImpl (Database Operations)
- DTO Layer: ElectricityBillDTO (Data Transfer Object)
- Database Layer: MySQL Database

6. Database Design

Database Name: electricity_db

Table: bills

Column Name	Data Type	Description
bill_id	INT (PK)	Bill ID
consumer_name	VARCHAR	Consumer Name
meter_no	VARCHAR	Meter Number
units	INT	Units Consumed
amount	DOUBLE	Bill Amount

7. Modules Description

7.1 Add Bill Module

This module accepts consumer name, meter number, and units consumed. The bill amount is calculated based on predefined slab rates and stored in the database.

7.2 View All Bills Module

This module retrieves and displays all electricity bill records from the database in a structured format.

7.3 Search Bill Module

This module searches and displays a bill using the meter number.

7.4 Update Bill Module

This module updates the units consumed and recalculates the bill amount for an existing meter number.

7.5 Delete Bill Module

This module deletes a bill record from the database using the meter number.

8. Bill Calculation Logic

- Units ≤ 100 → Units × 1.5
- Units ≤ 300 → 100 × 1.5 + (Units − 100) × 2.5
- Units > 300 → 100 × 1.5 + 200 × 2.5 + (Units − 300) × 4.0

9. Sample Output

- Add Bill Successfully
- View All Bills in Table Format
- Search Bill by Meter Number
- Update Bill Details
- Delete Bill Record

10. Advantages of the System

- Reduces manual calculation errors
- Fast and efficient bill management
- Easy to use console interface
- Secure database storage

11. Limitations

- Console-based interface
- Single-user access
- No online payment integration

12. Future Enhancements

- GUI-based application using Swing or JavaFX
- Login authentication system
- Online bill payment
- Report generation (PDF)

13. Conclusion

The Electricity Bill Management System successfully demonstrates the use of Java, JDBC, and MySQL to build a real-world application. The project helps in understanding database connectivity, layered architecture, and CRUD operations, making it suitable for academic purposes.

16. References

- Java Documentation
- MySQL Documentation
- JDBC API Reference